

Lab Activity

Mobile Sensor Networks – A.Y. 2025/26

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Content Outline

- **Wi-Fi and Bluetooth Lab (6h):**
 - Linux USB Stick boot + basic Wi-Fi scanning tools
 - Bluetooth management tools
 - Advanced Wi-Fi monitoring and trace analysis
- **Basic Network Programming Lab (3h):**
 - Basic networking programming with Python
 - Writing simple programs able to send and receive data in a Wi-Fi mobile network
- **Kubernetes Lab (9h):**
 - Lightweight virtualization technologies
 - Container orchestration systems
 - Practice on Kubernetes cluster with CrownLabs

Schedule

- Lab schedule is the **following, starting from May 7th**:
 - Tuesday 16:00 – 19:00 in Room 4I
 - Thursday 08:30 – 11:30 in Room 9T
1. May 7th → Lab introduction, basics on Wi-Fi and Bluetooth
 2. May 12th → Advanced Wi-Fi monitoring and trace analysis
 3. May 19th → Mobile network programming
 4. May 21st → Introduction to Kubernetes, Kubernetes lab
 5. May 26th → Kubernetes lab
 6. May 28th (*only two hours*) → Kubernetes lab
 7. June 4th → Kubernetes lab, wrap-up, discussion, doubts and thesis proposals

Rules

- The laboratory activity is mandatory and held in presence
- A virtual classroom is created for recording but in presence participation is strongly recommended
- Lab activity is composed of:
 - Basic theory and assignment introduction
 - Practical activity in groups
- Remember to bring your own laptop

Logistics

- You should have arranged in groups by yourself (4 people)

- **Final** group formation is available here:

https://docs.google.com/spreadsheets/d/1a4k6D6zAofSM9lghDGVb_RHmQhPTT97EfNAEEoPsgwE/edit?usp=sharing

The lab assignments will be uploaded on *Portale della Didattica*

- The lab assignments will guide you through the main activities

Reports

(Deadlines)

The reports must be delivered in **PDF** format

(see the next slides for details):

- Lab 1: Basic Wi-Fi scanning and Bluetooth tools (max 5 pages)
 - Deadline: May 18th
- Lab 2: Advanced Wi-Fi monitoring and trace analysis (max 5 pages)
 - Deadline: May 23rd
- Lab 3: Linux network programming (max 5 pages)
 - Deadline: June 3rd
- Lab 4: Kubernetes (max 5 pages)
 - Deadline: June 10th

Reports

(Delivery rules)

- Upload them on Portale della Didattica – “Elaborati” section
 - Deliverable: *lab_2026_<number>_group_<group_id>.pdf*
- For each lab activity, students have to describe the performed activity and obtained results, answer some questions and attach some screenshots, trying to go straight to the point
- **Expected format:**
 - Cover page (Group ID, members' names and student ID)
 - Describe the **activities** you carried out, your **results** and their **meaning**. Answer the questions explicitly and go directly to the point.
 - **Font 11 pt, single line spaced text**
 - Hard limit of 5 pages excluding the cover page
 - A template is provided in the Material folder

Reports

(Naming rules)

- Reports should be uploaded in PDF format (no .docx or .tex used to write the report), and should have the following name:
 - *Lab_2026_<number with a preceding "0">_group_<group_id>.pdf*
- Example for group 9 lab 04:
 - *Lab_2026_04_group_09.pdf*
- Example for group 15 lab 01:
 - *Lab_2026_01_group_15.pdf*
- Reports should follow the **expected format** (see the previous slide)
- Reports **not following** the expected format and/or with a **wrong name** are going to be graded with less points

Reports

(Evaluation)

- Please notice that: delivering all the reports is a basic requirement to pass the exam
- Delivery on time is mandatory
- The reports will be graded, and the grade will contribute to the final grade:
 - From 0 to 4 additional points
- The points gained through the lab activities will be valid till the winter exam session (January/February) included

How to write a good report

- You are writing a **scientific report** → you are expected to deliver a well-formated, easily readable PDF document
 - **Avoid** different fonts and sizes, images should have a **good quality, aspect ratio** and have a **caption**, plots should have **labels** to the x-y axis with **units** in square brackets (if you are plotting data with units) and a **legend** if applicable

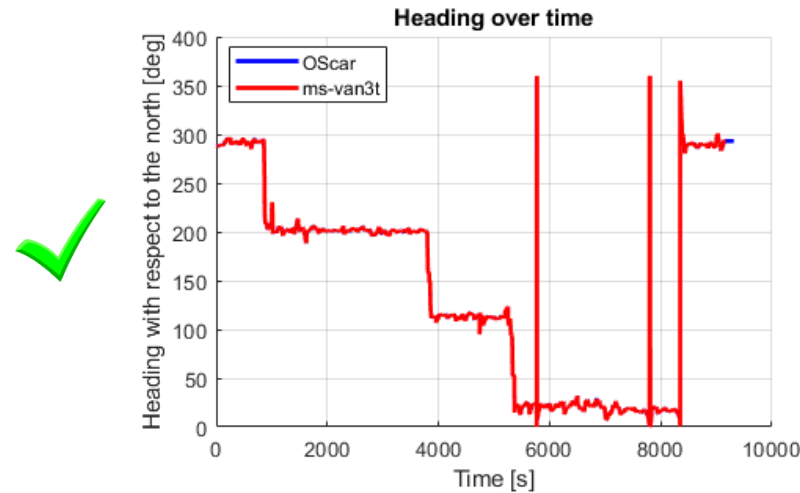


Figure 1. Heading over time as the vehicle travels along the test track.

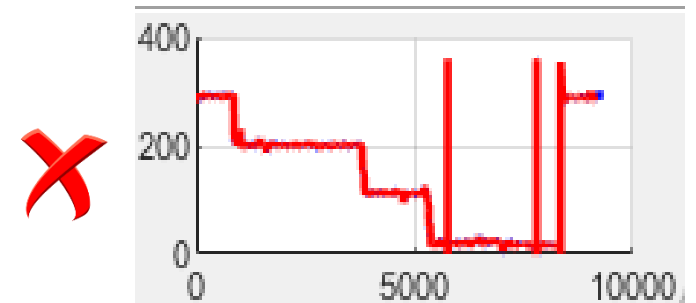


Figure 1. Heading.

How to write a good report

- **Avoid** long and generic introductions/conclusions, and write your own explanations/answers, going **directly to the point**
- **Remember** that you have a page limit of 5 pages, excluding the cover page
- Include **screenshots** and **images** when needed to better explain your results



The obtained results show how increasing the traffic priority from AC_BE to AC_VI causes an improvement in the measured one-way latency from 12 ms to 3 ms. This is due to the higher probability of transmitting before other nodes, thanks to the shorter AIFS time (79 μ s instead of 110 μ s) and contention window size.



In conclusion, this lab activity has strongly helped us understand the concepts at the base of mobile networks and has demonstrated the clear impact of traffic prioritization on latency. These findings highlight the importance of traffic management strategies in network performance, reinforcing the need to optimize prioritization for better efficiency in real-world applications. Furthermore, we learned how improving traffic priority can have an impact on how the network manages [...]

How to write a good report

- If you need to **report code** report the most important parts and **comment** them
- Report code/commands with a different **monospace** font (e.g. Consolas)
- **Always** check indentation and formatting when reporting **code** and **commands**



```
1  import ping3
2
3  for _ in range(5):
4      host = "8.8.8.8"
5      print("Pinging...")
6      ping3.ping(host)
```

The code reported above is used to ping the 8.8.8.8 remote server for 5 times. As can be seen, the ping3 library is imported (line 1), then [...] It is launched with `python3 ping.py`.



This is the code developed:

```
import ping3
for _ in range(5):
    host = "8.8.8.8"
    print("Pinging...")
    ping3.ping(host)
```

It is used to ping 8.8.8.8 and print something on the screen.

How to write a good report

- **AI** may be a powerful tool to help revising the text and make some activities faster and more efficient, if you know how to properly use it, but it **cannot** and **should not** do the lab for you!
 - The main goal of the lab activity is not (just) getting points at the exam but letting you learn, by practice and taking the time that you need, **new skills** that can be very important also for your future job after the master degree! Take time to **understand** what you are doing and **why** as you carry on the activity
 - It is **your report**, not the LLM report! Give your **own technical** explanations, with your **own words**, trying to explain what is asked starting from the knowledge that you have acquired during this and other courses, and through practical experience
 - The usage of AI tools for generating technical answers **can be detected**
 - It is often **clear** when specific answers have been generated with AI without a clear understanding of what has been done → this is going to be **strongly penalized**



How to write a good report

- Check **grammar** and **typos**



After setting the coloring rules as described earlier, the first and last packet of the Bluetooth Low Energy (BLE) file transfer can be easily identified at time 5 s and 7.5 s.



In setting the coloring rules **since** described earlier, the first and last packet of the **Bluetooth** Low Energy (BLE) file **trasnfer** can be easily identified at time 5 s and 7.5 s.

- The report **evaluation** will also take the **document quality** into account

Prerequisites

(Wi-Fi and Bluetooth)

All students need to satisfy the following requisites before the lab:

- Wi-Fi at home or at university (e.g., *polito*, *eduroam*). Any ordinary access point is enough.
- A laptop (or PC with Wi-Fi) running Ubuntu 22.04. If you don't have it, you can download a live image and run it from a USB stick. Instructions on how to build a bootable USB stick with Ubuntu can be found in the following link:
 - <https://ubuntu.com/tutorials/tutorial-create-a-usb-stick-on-windows>
- For some experiments, you will need a second device connected to your home/university Wi-Fi network, e.g., a laptop, PC, Android/iPhone

Prerequisites

(Linux network programming)

All students need to satisfy the following requisites before the lab:

- Wi-Fi at home or at university (e.g., *polito*, *eduroam*). Any ordinary access point is enough.
- A laptop (or PC with Wi-Fi) running Ubuntu 22.04. If you don't have it, you can download a live image and run it from a USB stick. Instructions on how to build a bootable USB stick with Ubuntu can be found in the following link:
 - <https://ubuntu.com/tutorials/tutorial-create-a-usb-stick-on-windows>
- You will also need a second laptop or device connected to the same Wi-Fi network (you can take two laptops between the ones of the group members) → prepare if possible two bootable USB sticks if you don't have Linux on two devices

Prerequisites

(Kubernetes)

- The activity leverages *kubeadm* tool for the installation and configuration of a Kubernetes cluster
 - Each group will instantiate a set of Virtual Machines (VMs)
 - These VMs will be hosted on **CrownLabsNG@POLITO**
(<https://ng.crownlabs.polito.it/>)
- Thus, the only requisite is to bring your own laptop to class
- You will be added to a CrownLabs workspace before the beginning of Lab 4, that will start on May 19th



Questions?

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